

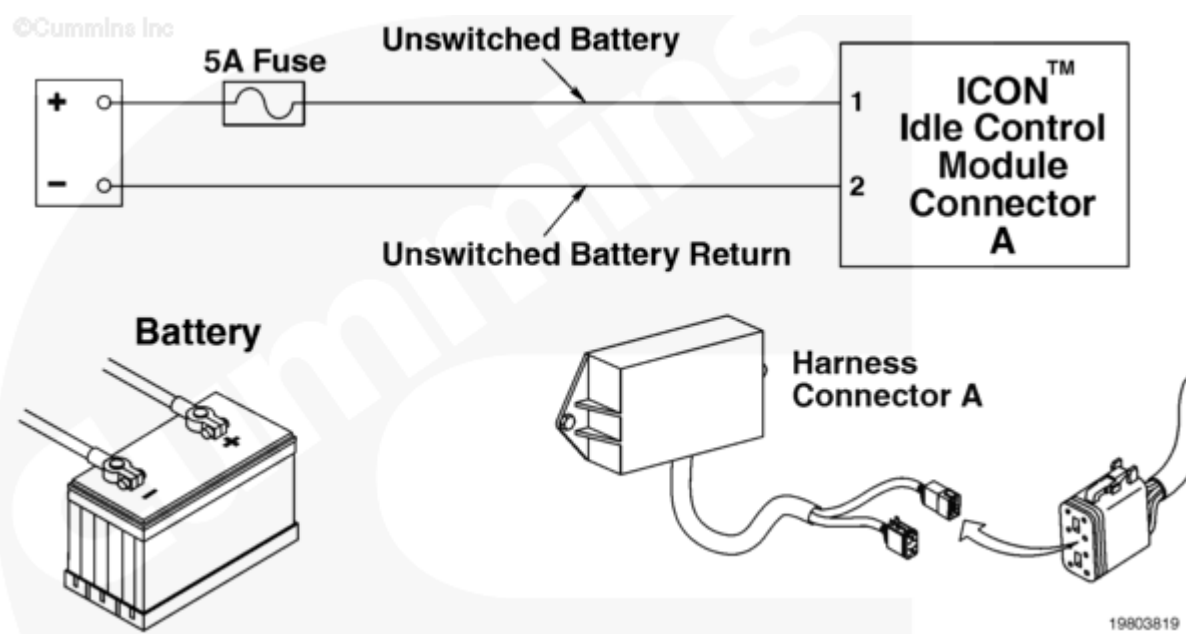
Fault Code: 441 (Aftermarket and OEM)

Battery Number One Voltage Low - Data Valid But Below Normal Operational Range - Moderately Severe Level

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 441 PID(P): SPN: FMI: Lamp: SRT:	Battery Number One Voltage Low - Data Valid But Below Normal Operational Range - Moderately Severe Level. Battery voltage below normal operating level.	The ICON™ system will be disabled. Only mandatory shutdown will be enabled. Engine will start normally.



Circuit Description

The ICON™ idle control module receives unswitched battery input through the ICON™ engine harness. There is one 5-amp in-line fuse in the unswitched battery wire to protect the ICON™ engine harness wire. The battery return wires are connected directly to the negative (-) battery post. The above circuit diagram can vary, such as connector or pins, depending on the vehicle make or model. OEM installations can possibly provide the harnessing between the idle control module and other ICON™ devices.

Component Location

The ICON™ idle control module is connected directly to the batteries through the ICON™ engine harness. This direct link provides a constant power supply for the ICON™ idle control module. Refer to the OEM manual for the battery location. The ICON™ module can be located in a different location depending on the vehicle application.

Shoptalk

The unswitched battery supply and return wires **must** be directly connected to the battery for the ICON™ system to function properly. During the ICON™ system start, this fault can be logged during engine cranking if there is a faulty ground connection.

This fault will be logged if the battery voltage falls below 9 VDC on a 12 VDC system. This is equivalent to a very low battery voltage on the engine electronic control module (ECM).

Perform a battery charging system test as described in Procedure 210-001 (</qs3/pubsys2/xml/en/procedures/97/97-210-001.html>), Installation Guidelines, to verify the battery will have adequate voltage for the ICON™ system to function properly.

The ICON™ system can display **only** the present active fault code. If more than one fault code is active at the same time, the ICON™ system flashes out the highest priority fault. After the fault has been corrected then the next active fault will be flashed out.

Note: The ICON™ electronic service tool can display more than one active and or inactive fault codes at the same time.

Warnings and Cautions

CAUTION

To reduce the possibility of pin and harness damage, use the following test lead when taking a measurement:

Part Number 3822758 - male Deutsch/AMP/Metri-Pack test lead.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the batteries and charging system.	
	STEP 1A. Check the batteries and alternator.	No damaged connections
STEP 2.	Check the fault status.	

STEPS		SPECIFICATIONS
	STEP 2A. Use the fault flashout feature or the ICON™ electronic service tool to read the fault codes.	Fault Code 441 inactive
STEP 3.	Clear the fault code.	
	STEP 3A. Disable the fault code.	Fault Code 441 cleared

STEP 1. Check the batteries and charging system.

STEP 1A. Check the batteries and alternator.

Conditions : Turn keyswitch OFF.		
Action	Specification/Repair	Next Step

- For M11 engines, refer to Procedure 013-001 (/qs3/pubsys2/xml/en/procedures/36/36-013-001.html), Procedure 013-007 (/qs3/pubsys2/xml/en/procedures/36/36-013-007.html), and Procedure 013-009 (/qs3/pubsys2/xml/en/procedures/36/36-013-009.html) in the base engine Troubleshooting and Repair Manual, Bulletin 3666139.
- For N14 engines, refer to Procedure 013-001 (/qs3/pubsys2/xml/en/procedures/09/09-013-001.html), Procedure 013-007 (/qs3/pubsys2/xml/en/procedures/09/09-013-007.html), and Procedure 013-009 (/qs3/pubsys2/xml/en/procedures/09/09-013-009.html) in the base engine Troubleshooting and Repair Manual, Bulletin 3666142.
- For ISM engines, refer to Procedure 013-001 (/qs3/pubsys2/xml/en/procedures/35/35-013-001.html), Procedure 013-007 (/qs3/pubsys2/xml/en/procedures/35/35-013-007.html), and Procedure 013-009 (/qs3/pubsys2/xml/en/procedures/35/35-013-

More than 12 VDC

2A

009.html) in the base engine Troubleshooting and Repair Manual, Bulletin 3666322.

- For Signature and ISX engines, refer to Procedure 013-001 (/qs3/pubsys2/xml/en/procedures/10/10-013-001.html), Procedure 013-007 (/qs3/pubsys2/xml/en/procedures/10/10-013-007.html), and Procedure 013-009 (/qs3/pubsys2/xml/en/procedures/10/10-013-009.html) in the base engine Troubleshooting and Repair Manual, Bulletin 3666239.

Troubleshoot any other fault codes.

3A

STEP 2. Check the fault status.

STEP 2A. Use the fault flashout feature or the ICON™ electronic service tool to read the fault codes.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect the ICON™ electronic service tool.

Action	Specification/Repair	Next Step
	Fault Code 441 inactive	3A
	Fault Code 441 active Replace the ICON™ idle control module. Refer to Procedure 019-358 (/qs3/pubsys2/xml/en/procedures/97/97-019-358.html).	Repair Complete

STEP 3. Clear the fault code.

STEP 3A. Disable the fault code.

Conditions :

- Connect all components.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
• Cycle the keyswitch to verify the fault code is inactive.	Fault Code 441 cleared	Repair complete
	Troubleshoot any remaining active fault codes.	Appropriate troubleshooting charts

